

Submission: Bring art to life with AR

Shows to watch

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PATHWAY

Mobile AR Development

Unity Version: 2022.3

Contents

TUTORIAL

EXERCISE

Submission: Bring art to life with AR

1 • Overview

2 • Project brief: Bring art to life with AR

3 • Requirements

4 • Additional submission guidance

5 • Submit your work

MISSION

Create a plane detection AR app

MISSION

AR experience design

Pathway completion details

Viewable by:

Public (Anyone who views this tutorial)

Private (Just me for now)

Save and preview submission

Exercise complete!

+20 XP

You gained 20 XP points towards Art & Animation

+20 XP

You gained 20 XP points towards Systems Design

+20 XP

You gained 20 XP points towards Programming

You completed this exercise on March 07, 2026

Next up in Mobile AR Development

5

Create a plane detection AR app

+300 XP • 8 Hours 5 Mins

Continue to next

Submission Gallery

Shows to watch

How To: Use Fresh

Hands-on Activity

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BULAONG, ROSS ANDREW

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NEWACTIVITY ↑

REALITY

EXTENSIVE EXTENDED REALITY

24 PTS

DUE: 11:59 PM

Creating a Marker-based AR

HISTORY

Planned Yet

CALENDAR EVENT

ION

ALL DAY

IGN

Submitted

Mobile AR Development - Unity

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Last visited on March 07, 2026

56% complete

MISSION

Get started with AR

MISSION

Create a basic face filter

MISSION • COMPLETED MARCH 05, 2026

Create an interactive face filter

MISSION • COMPLETED MARCH 07, 2026

Create a marker-based AR app

MISSION

Create a plane detection AR app

8 Hours 5 Mins

Make an app that detects flat surfaces and allows the user to spawn a portal on their floor or wall. Then you'll create an interactive character app.

Mission Details

Tutorial • Set up a scene with plane detection +10 XP

Tutorial • Detect the user's touch +10 XP

Tutorial • Detect when the user touches a plane +10 XP

Tutorial • Instantiate the portal on a plane +10 XP

Tutorial • Test the frame rate of the portal world +10 XP

Tutorial • Limit the visibility of the portal +10 XP

Joy Horvath

Unity Learn Pathway's Host

Level:

Foundational

Pathway completion bonus:

+800 XP

Completion time:

8 weeks

Pathway Description

Welcome to AR Development! In this learning Pathway, you'll create Augmented Reality (AR) applications that are compatible with iOS and Android devices. You'll use Visual Scripting and Unity's AR tools to develop a wide variety of engaging AR experiences. This pathway assumes a basic knowledge of Unity.

Skills you will learn

Beginner Mobile Optimization

Absolute Beginner AR Design

Absolute Beginner Visual Scripting

Shows to watch

How To: Use Fresh

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Submitted

The screenshot displays a Mac desktop with two primary applications open. On the left, the Unity 2022.3.62f3 development environment is running the 'ARMarker - Esguerra_ARProject - iOS' project. The interface includes a Hierarchy panel on the left showing the scene's structure, a central 3D viewport with a yellow cat-like character, and a bottom panel with the Project, Console, and Visual Scripting Graph tabs. The Project panel shows a hierarchy of assets including Animations, Fonts, Markers, Materials, Prefabs, Sounds, SourceFiles, UI, VisualScripts, ARMarker, MobileARTemplateAssets, Samples, Scenes, Settings, and TaxiMash Pro. The Visual Scripting Graph tab is active, showing a collection of markers and assets like 'Business...', 'Markeric...', 'possum', and 'Referenc...'. On the right, a web browser window is open to the profile page of 'ESGUERRA, SERGE' on the 'tip.instructure.com' website. The profile page features a circular profile picture, an 'Edit Profile' button, and sections for 'User Profile', 'Contact' (noting no registered services), and 'Biography' (noting no biography has been added). The browser's address bar shows the URL 'tip.instructure.com/profile'. The Mac's dock at the bottom contains various application icons, including Finder, Launchpad, Safari, Messages (with 187 notifications), Mail, Photos, Video, Calendar (showing 'Thu 5'), Music, App Store, System Settings (with 1 notification), Phone, Chrome, and several other utility and development tools.

PATHWAY
Mobile AR
Development

Unity Version:
2022.3

Contents

MISSION
Create a marker-based AR app

TUTORIAL
Create a new project for marker-based AR

TUTORIAL
Spawn an object on an AR Marker

TUTORIAL
Add a UI with rotation buttons

TUTORIAL
Add world space UI to your marker-based app

TUTORIAL
Add spatial audio

Exercise complete!

+20 XP

You gained 20 XP points towards Art & Animation

+20 XP

You gained 20 XP points towards Systems Design

+20 XP

You gained 20 XP points towards Programming

You completed this exercise on March 07, 2026

Next up in Mobile AR Development

5

Create a plane detection AR app

+300 XP • 8 Hours 5 Mins

Continue to next

User Profile

Edit Profile

Edit Profile Picture

ESGUERRA, SERGE

Contact

No registered services, you can add some on the [settings](#) page.

Biography

No biography has been added

PATHWAY
Mobile AR Development

Version: 2022.3

Contents

TUTORIAL
Spawn an object on an AR Marker

TUTORIAL
Add a UI with rotation buttons

TUTORIAL
Add world space UI to your marker-based app

TUTORIAL
Add spatial audio to your marker-based app

EXERCISE
Submission: Bring art to life with AR

What you will learn

A **marker-based** app is an application that can recognize images (called markers) in the real world and then make content appear in the app. You may have seen applications that use this kind of technology to make AR content for movie posters, product labels, card games, and even business cards.

In this project, you'll make a marker-based app with a user interface that lets the user control a 3D model, text, and audio.

Skills you will learn

Beginner Mobile Optimization

Additional XP earned by completing the unit

+300

You completed this and earned the XP on on Mar 07, 2026

User Profile

Edit Profile

Edit Profile Picture

ESGUERRA, SERGE

Contact

No registered services, you can add some on the [settings](#) page.

Biography

No biography has been added

PALOMARES

The image is a collage of three screenshots from a digital workspace.

Top Left Screenshot: A web browser window showing a user profile for "PALOMARES, CHARLES YEXEL" on the "tip.instructure.com" dashboard. The profile includes a "Logout" button, a sidebar menu with options like "Dashboard", "Courses", "Groups", "Calendar", "Inbox", "History", and "Help", and a list of "Accessibility Settings" such as "Use High Contrast UI" and "Use a Dyslexia Friendly Font".

Top Right Screenshot: A Unity 2022.3.623 (64-bit) interface for creating a marker-based AR application. The Hierarchy panel shows a "XR Origin (AR)" object with a "XR Session" and "XR Interaction" component. The Inspector panel shows the "XR Interaction" component with a "buttonRotateLeft" event. The Console panel shows a "Script Graph" with a "Get Variable" node and a "buttonRotateLeft" event.

Bottom Screenshot: A submission completion screen for a "Mobile AR Development" pathway. The screen shows a "Submission: Bring art to life with AR" and a "Exercise complete!" message. It lists three completed exercises: "Add world space UI to your marker-based app", "Add spatial audio to your marker-based app", and "Bring art to life with AR". The total XP gained is 60, and the completion date is March 05, 2026. The screen also shows a "Next up in Mobile AR Development" section with a "Create a plane detection AR app" mission and a "Submission Gallery" section.

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Tip-Instructure

PALOMARES, CHARLES YEXEL

as a member of the T.I.P. Community.

ing the information contained in this

issued memoranda.

ork on Wednesday, February 25, 2026 in

sary of the EDSA People Power Revolution

of the Philippines

orking) Day in T.I.P. Manila and T.I.P. Quezon City in

Mobile AR Development - Unity

Hands-on Activity 3.2 - Creatin

learn.unity.com/pathway/mobile-ar-development?version=2022.3

Ready to create AR experiences? In this learning pathway, you'll develop AR apps compatible with iOS and Android devices!

Continue Pathway

Last visited on March 05, 2026

Learners that completed pathway: 1517

Joy Horvath

Unity Learn Pathway's Host

Level: Foundational

Pathway completion bonus: +600 XP

Completion time: 8 weeks

Pathway Description

Welcome to AR Development! In this learning Pathway, you'll create Augmented Reality (AR) applications that are compatible with iOS and Android devices. You'll use Visual Scripting and Unity's AR tools to develop a wide variety of engaging AR experiences. This pathway assumes a basic knowledge of Unity.

Skills you will learn

Content Outline

Unity Version 2022.3

56% complete

MISSION 1 Get started with AR

MISSION 2 Create a basic face filter

MISSION 3 Create an interactive face filter

MISSION 4 Create a marker-based AR app

5 Hours 15 Mins

Learn to create marker-based AR apps, which make digital models appear over marker images in the real world.

Mission Details

Tutorial Create a new project for marker-based AR +10 XP

Tutorial Spawn an object on an AR Marker +10 XP

Tutorial Add a UI with rotation buttons +10 XP

Tutorial Add world space UI to your marker-based app +10 XP

Tutorial Add spatial audio to your marker-based app +10 XP

Exercise Submission: Bring art to life with AR +60 XP

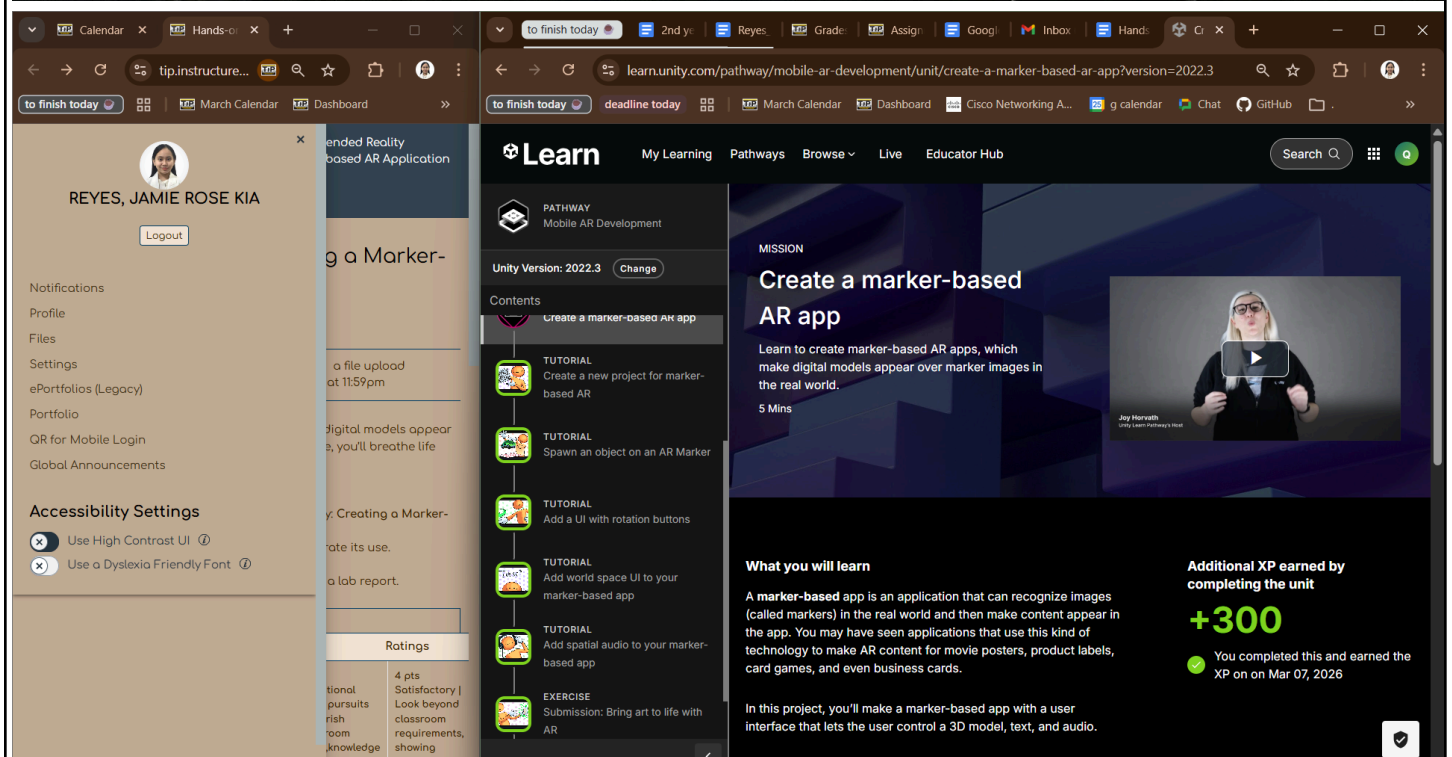
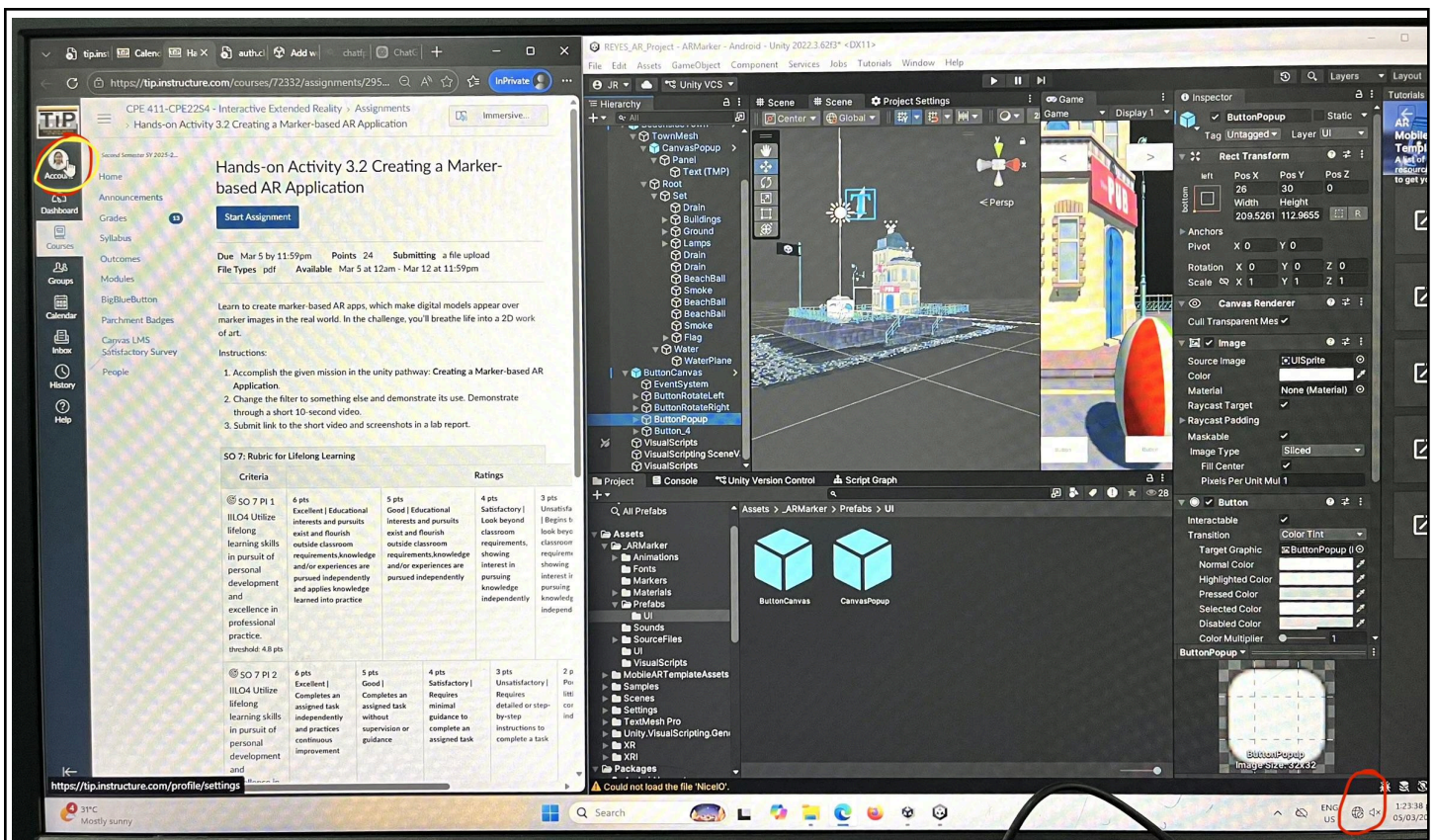
23°C

Partly cloudy

Search

6:38 pm

07/03/2026



4. Conclusion

Bulaong, Ross Andrew D.:

In this activity, I was able to create a marker-based AR application that is able to view and rotate a scared lemon-like humanoid character named John Lemon in real life. John Lemon appears after the app detects a "marker" (e.g., a printed image) set by using an AR Reference Image in the Unity project. During the development process, I encountered several issues such as the 3D model not appearing correctly, rotation scripts not functioning as expected, and user interface buttons not responding. These challenges

required me to carefully review the steps in the tutorial and troubleshoot possible errors in the setup of the AR components and scripts. After revisiting the steps and testing different solutions, I was able to successfully fix the problems and produce a working marker-based AR application.

Esguerra, Serge

In conclusion, in this activity, I followed all the steps of developing a marker-based AR application. During this activity, I encountered many errors, such as the 3D model not rotating or not showing, and the buttons not working. At first, this was a challenge, I redo all the steps from the start and tried different kinds of solutions to fix the problem. Through this, I have been able to overcome the challenges. At the end of the exercise, I developed a working marker-based AR application and also improved the design by customizing the user interface. Additionally, I have developed another marker-based AR application using a cat as a 3D model and changed the text and audio. Overall, I have gained a better understanding of the creation of marker-based AR applications. This has been a great learning experience for me, and improved my skills in the creation of AR applications.

Palomares, Charles Yexel

Learning this activity is fun, I make marker-based AR applications. First, I have many errors—3D models not showing and buttons not working. It is very hard, so I restart and try different ways to fix it. Finally, I solved everything and made the app work and I was able to see John-Lemon. I even make the UI look workable! I also tried another app with a cat model and new audio. Now, I understand AR well enough. This is a great experience and my skills are very much improving. Thus, I will try to practice more to be able to create my own AR application.

Reyes, Jamie Rose

In this activity, creating a marker-based AR app was really challenging for me because I had to deal with many new and unfamiliar problems. I struggled a lot with the beachside town 3D model that I chose to appear on my AR marker, especially during the rotation tutorial where only the beach part was rotating while the rest of the town stayed still. It was frustrating at first and took me some time to figure out what was causing the issue. Even though the process was difficult and very time-consuming, working through these problems helped me understand more about how AR works. In the end, I'm still glad that I got to study this field because finishing the app felt really rewarding and made all the effort worth it.

Video Link:

<https://drive.google.com/drive/folders/1GWMsimiO2PuElloAnGLEPI98FWDBOPWq?usp=sharing>

6. Assessment Rubric

Lab Activity Rubric



Criteria	Ratings						Pts
 SO 7 PI 1 Student Outcome 7.1 Acquire and apply new knowledge from outside sources. threshold: 4.8 pts	6 pts Excellent Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently	4 pts Satisfactory Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor Relies on classroom instruction only	1 pts Very Poor No initiative or interest in acquiring new knowledge	6 pts
 SO 7 PI 2 Student Outcome 7.2 Learn independently threshold: 4.8 pts	6 pts Excellent Completes an assigned task independently and practices continuous improvement	5 pts Good Completes an assigned task without supervision or guidance	4 pts Satisfactory Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory Requires detailed or step-by-step instructions to complete a task	2 pts Poor Shows little interest to complete a task independently	1 pts Very Poor No interest to complete a task independently	6 pts
 SO 7 PI 3 Student Outcome 7.3 Critical thinking in the broadest context of technological change threshold: 4.8 pts	6 pts Excellent Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory Apply the gathered information to formulate the problem	2 pts Poor Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor Gather information from a variety of sources	6 pts
 SO 7 PI 4 Student Outcome 7.4 Creativity and adaptability to new and emerging technologies threshold: 4.8 pts	6 pts Excellent Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory Shows some creative ways to solve the problem	2 pts Poor Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor Ideas are copied or restated from the sources consulted	6 pts
Total Points: 24							